Algorithm

- **Step-by-Step Procedure**

1. **Input Text**
    - Accept user input (sentence or word).
2. **Preprocessing**
    - Convert text to lowercase
    - Remove punctuation
    - Tokenize words
3. **Error Detection**
    - Compare each word with a dictionary
    - Identify misspelled words
4. **Candidate Generation**
    - Generate possible corrections using:
        - Insertions
        - Deletions
        - Substitutions
        - Transpositions
5. **Probability Calculation**
    - Use word frequency to rank suggestions
    - Choose the most probable word
6. **Correction Output**
    - Replace incorrect words
    - Display corrected sentence

- **Logic / Method Used**

- **Edit Distance (Levenshtein Distance)**
- **Probability-based correction (word frequency)**
- **Dictionary lookup**

- **NLP techniques**

🔹 **Model / Technique Applied**

- Rule-based approach
- Statistical NLP model
- Optional: N-gram language model

🔹 **Process Flow**

User Input → Preprocessing → Error Detection → Candidate Generation
→ Probability Selection → Corrected Output

Datasets

🔹 **Dataset Description**

The dataset consists of a large collection of correctly spelled words along with their frequency of occurrence.

🔹 **Types of Dataset Used**

1. **Dictionary Dataset**
    - List of valid English words
    - Example: word list file (words.txt)
2. **Corpus Dataset**
    - Large text data (books, articles)
    - Used to calculate word frequency

🔹 **Example Dataset**

the 50000
hello 3000
world 25000
python 2000
project 150